

AARAV KHANAL

Bengaluru, India | khanalaarav2063@gmail.com | +91 9902835677 | [LinkedIn](#) | [GitHub](#)

SUMMARY

AI & Machine Learning undergraduate seeking a Software Engineering / AI Engineering internship. Builds full-stack, multi-agent AI systems end-to-end — from LLM orchestration and model routing to deployment — across four independent production-style platforms spanning developer tooling, decision intelligence, AutoML, and computer vision.

TECHNICAL SKILLS

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks

Languages: Python, C++, JavaScript, TypeScript, SQL

AI/ML: LLM Orchestration, Multi-Agent Systems, RAG, Agentic AI, AutoML, Explainable AI (SHAP/LIME), Computer Vision, NLP

Frameworks/Libraries: PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, FastAPI, React, Next.js, TailwindCSS

Infrastructure: Docker, Git, Linux, PostgreSQL, MongoDB, SQLite, WebSockets, REST APIs

LLM Providers: OpenAI, Anthropic, Gemini, NVIDIA NIM, Groq, Ollama, LM Studio

ENGINEERING EXPERIENCE

Independent AI Engineer — ProjectMind AI

2025 – Present

Python, LangGraph, SQLite | github.com/Aaravkhanal/ProjectMind-AI

- Integrated 10+ LLM providers into a BYOK (bring-your-own-key) multi-agent code review platform, with an environment-discovery engine that auto-detects a developer's IDE, API keys, local models, and MCP servers on startup.
- Designed a persistent "Repository Brain" (SQLite) that indexes code hotspots, technical debt, and contributor history across sessions, giving every review long-term project context instead of stateless one-off analysis.
- Orchestrated 3 parallel review agents (Architecture, Security, Quality) via LangGraph, feeding a Synthesizer agent that deduplicates and prioritizes findings into a single actionable verdict.
- Implemented an intelligent model router with cost-aware selection logic, automatically matching each task type to the best available model instead of relying on manual model choice.

Independent AI Engineer — JUBIDATE

2025 – Present

Next.js, FastAPI, WebSockets | github.com/Aaravkhanal/Jubidate_ai

- Coordinated 3 specialized LLM agents (Pro Advocate, Con Advocate, Judge) in a full-stack orchestration platform that debates and adjudicates complex strategic questions in real time.
- Streamed concurrent, turn-based generation across 6+ LLMs (Llama 3.1 70B, Nemotron, Mixtral via NVIDIA NIM, plus Gemini, Groq, and OpenRouter) over a low-latency WebSocket pipeline.
- Shipped and deployed both frontend (Vercel) and backend (Railway) with a documented Swagger API, taking the project from design through live production deployment independently.

Independent AI Engineer — NeuralForge AI

2025 – Present

FastAPI, LangGraph, Docker | github.com/Aaravkhanal/NEURALFORGEAI

- Coordinated 10 specialized LLM agents (Data Scientist, ML Engineer, DevOps, etc.) to guide a user from raw data through automated cleaning, model training, and deployment.
- Built an AutoML pipeline benchmarking 3+ model families (XGBoost, Random Forest, LightGBM) and automatically selecting the best performer.
- Implemented an Explainable AI playground (SHAP, LIME, decision-path visualization) with a natural-language "AI Teacher" mode that explains model predictions in plain English.
- Automated model export across 7+ formats (ONNX, joblib, PMML) and generated one-click deployment packages (FastAPI, Docker, Streamlit).

Independent AI Engineer — AURA

2025 – Present

Node.js, IDM-VTON, Pinterest API | github.com/Aaravkhanal/fashion_ai

- Integrated the Pinterest Data API with Hugging Face's IDM-VTON diffusion model to power an AI-driven virtual try-on ecosystem with real-time trend discovery.
- Designed a biometric-based digital-twinning pipeline (height, weight, face-scan input) to personalize garment fit and rendering for each user.

EDUCATION

B.M.S. College of Engineering, Bengaluru

B.E. Artificial Intelligence & Machine Learning

CGPA: 8.0/10 | Expected Graduation: 2028

Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Machine Learning, Deep Learning, Computer Vision, NLP, Software Engineering

ACTIVITIES

- Participant in multiple university-level AI and software engineering hackathons.
- Independently researches and builds in multi-agent systems, LLM orchestration, and generative AI.